Modeling Volcanic Flows: Lahars

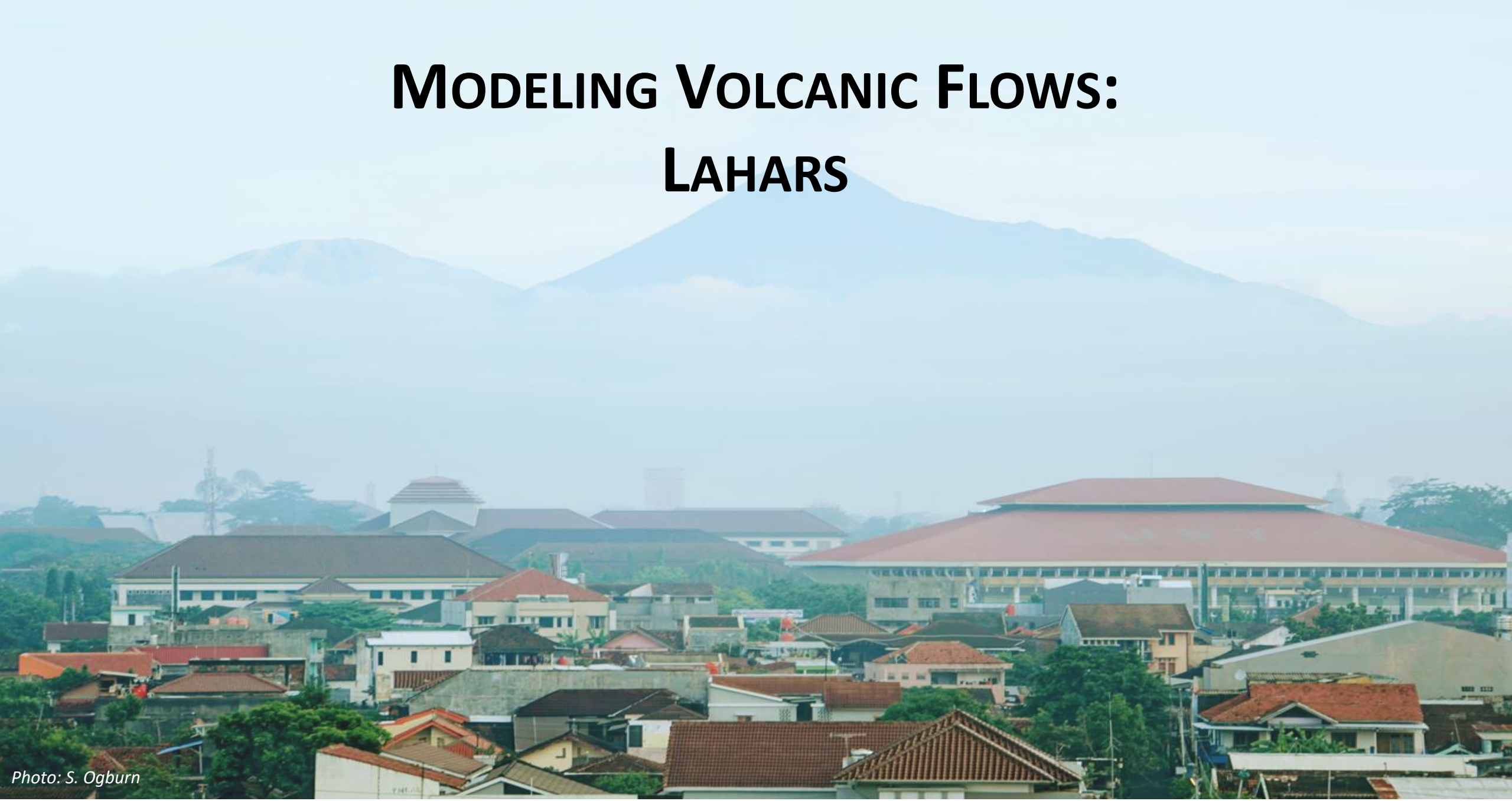


Photo: S. Ogburn

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Lahar/debris flow models

- LAHARZ (Iverson et al. 1998): empirical model based on flow volume, inundated area, and maximum cross-sectional area
- Flo2D (OBrien 1993): hydrodynamic model
- TITAN2F (Cordoba et al. 2015): new two-phase flow version of TITAN2D
- VolcFlow (Kelfoun & Druitt 2005): single and two-phase flow models
- D-CLAW (George & Iverson 2014): two phase debris flow model with pore-fluid pressure, evolving volume fractions, etc.

Model	Citation	Description/notes	Availability	Count	Percent
LAHARZ (USGS)	Iverson et al. (1998); Schilling (1998); Schilling (2014)	Empirical/statistical model relating volume to planimetric/cross-sectional area	Free, open-source ArcGIS, standalone python, VICTOR	149	71.2
FLO-2D	O'Brien (1993)	2D depth averaged geophysical model; Bingham rheology; designed by engineers for floods, floods in urban environments	Commercial product license $	23	11.0
LaharFlow/MultiHaz	Woodhouse (in preparation); Tierz et al. (2017)	Depth-averaged geophysical model; two-phase, models erosion and deposition	In preparation	6	2.9
HIMAT-FAO	HIMAT-FAO (1987)	?	?	4	1.9
MatHaz	Bertin et al. (2019)	Multi-hazard model; steepest descent/volume-thickness model developed for lava flows, modified for lahars	Free, open-source, available on GitHub	3	1.4
DAMBRK/SMPDBK	Fread (1984); Wetmore & Fread (1991)	Designed for dam-break floods from failed reservoir; models flood peak elevation, travel time using flow routing algorithm	Free, open-source, unclear if still supported	2	0.9
HYDRAUX (USGS)	Hansen (1987); DeLong & Schoellhamer (1989)	1D shallow water geophysical flow model	FORTRAN, open-source, no longer available/supported	2	0.9
VolcFlow	Kelfoun & Druitt (2005)	Geophysical mass flow model developed for granular flows but with multiple rheology options for lahars, one-fluid and two-fluid options	Free, open-source but requires Matlab	2	0.9
Unnamed cost-distance model	Barrantes Castillo & Malavassi Rojas (2015)	GIS-based cost-distance model that determines potential flow paths	Not available, but could implement method in any GIS	1	0.48
Unnamed lahar model	Miyamoto & Ito (2003)	?	?	1	0.48
Unnamed lahar travel time model	Pierson (1998)	Empirical lahar travel time model, does not model inundation	Equations relating flow volume, distance, and time available	1	00.48
Unnamed mass flow model	Mastrolorenzo & Pappalardo (2010)	Geophysical mass flow model developed for granular flows but with multiple rheology options for lahars	Not available	1	0.48

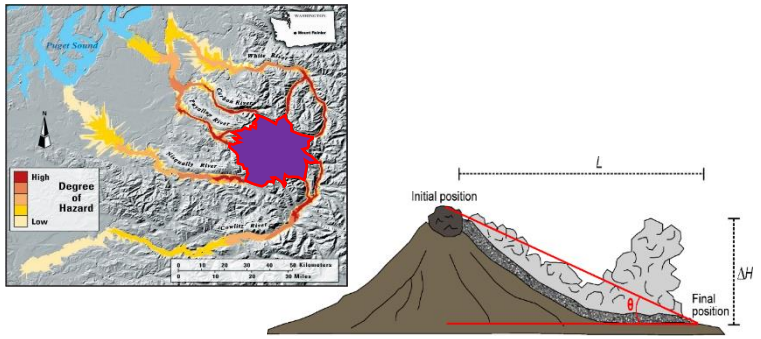
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Modeling volcanic flows

- “All models are wrong, but some are useful” –statistician George Box
- Different models are wrong in different ways (make different simplifications and assumptions)
- Different models might be more wrong (or more useful) for certain questions, types of hazards, or situations
  - One model might be better for simulating large PDCs than small PDCs, or one model might be better at simulating velocities while another is better at simulating inundation
  - One model might be better if you only have a short amount of time, or if you have a poor quality DEM

Increasing complexity and computational time

Statistics-based (Empirical)



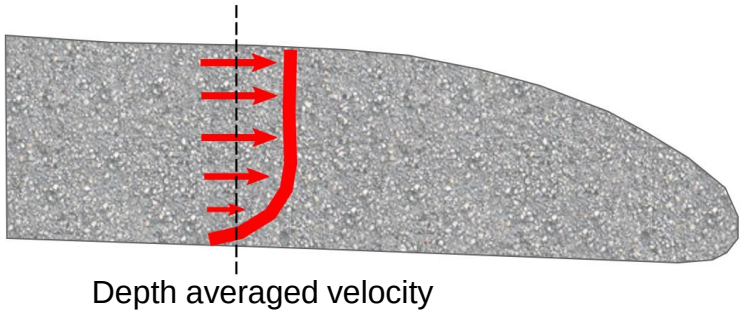
Empirical models

-Based on local and global data

-Runout or inundated areas derived from statistical relationships with flow volume

-Ex: Energy cone (Malin & Sheridan, 1982), FLOW2D (Sheridan & Macias, 1992), LAHARZ (Iverson et al. 1998, Schilling et al. 2014)

Physics-based

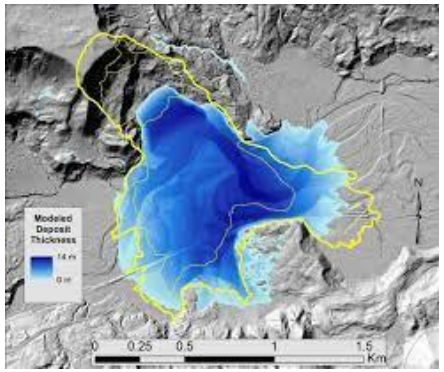


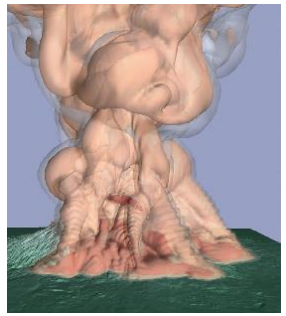
Depth-averaged models

-Relatively thin, dense, concentrated flows

-Frictional interaction at the bed or constant retarding shear stress controls flow motion

-Ex: TITAN2D (Patra et al. 2005), VolcFlow (Kelfoun & Druitt , 2005)





Multiphase models

-Separate conservation equations for multiple particle types

-Drag between the gas and particles transmits momentum between phases

-Ex: PDAC (Ongaro et al. 2008), (Dufek & Bergantz 2007), DCLAW (George et al. 2014

An example of different models

LAHARZ (statistical)

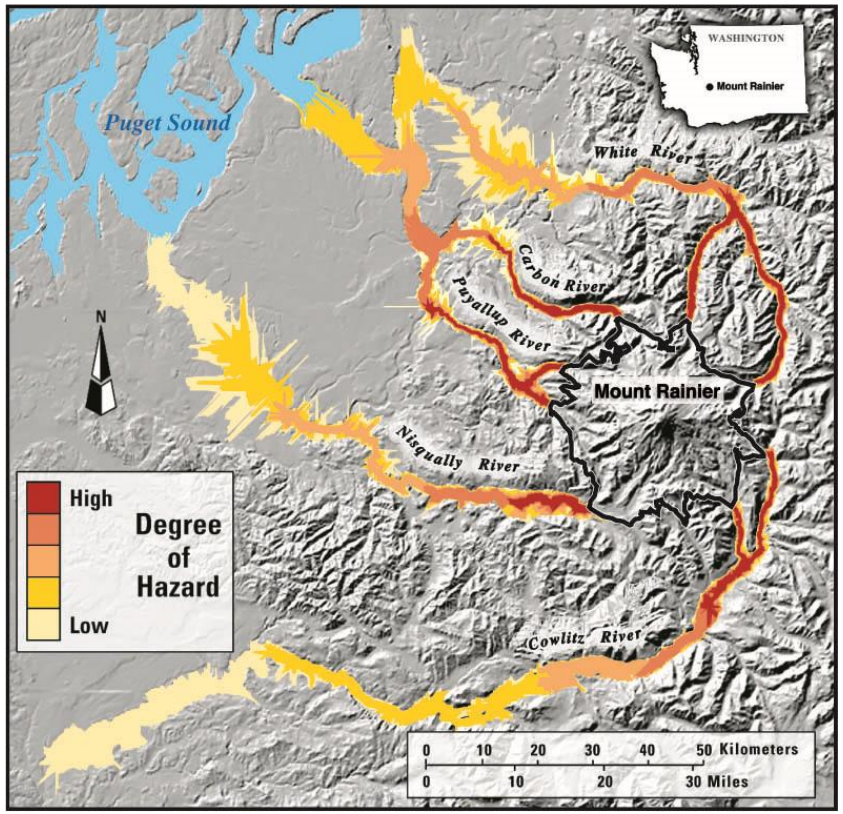
- Empirically based
- Few model input parameters (DEM, stream network, start locations, volumes)
- Vary volumes
- Gives simple maps of probable lahar inundation for a given volume
- Quick, not much computing power

D-CLAW (physical)

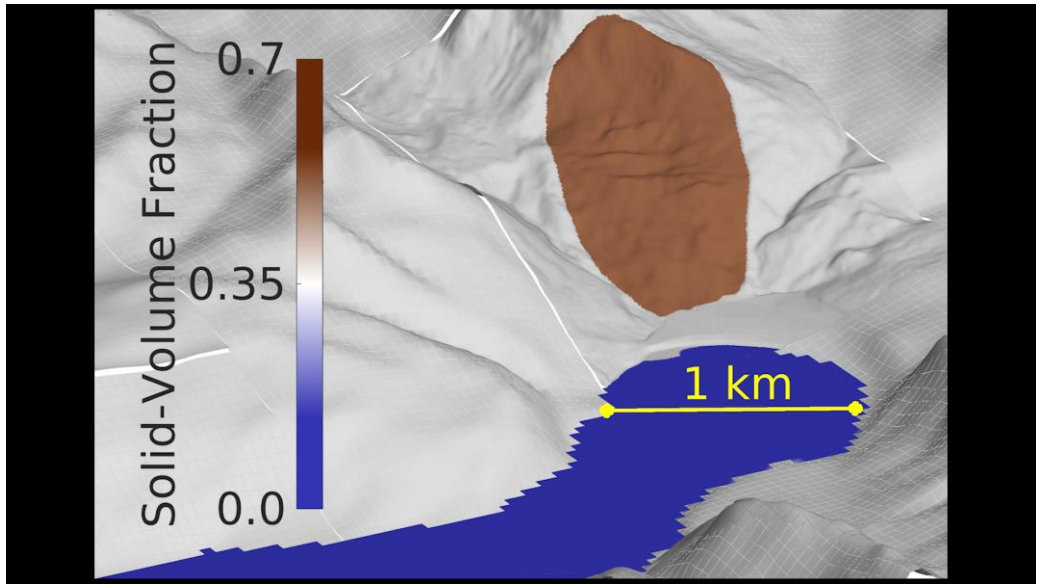
- Physical flow model
- Many model input parameters (DEM, precise 3D source area, 8 different material properties)
- Input parameters are very difficult to know ahead of time so must be varied with many simulations
- Simulates dynamics of flows and gives inundation, velocity, pore pressure evolution
- Computationally demanding

An example of different models

LAHARZ (statistical)



D-CLAW (physical)



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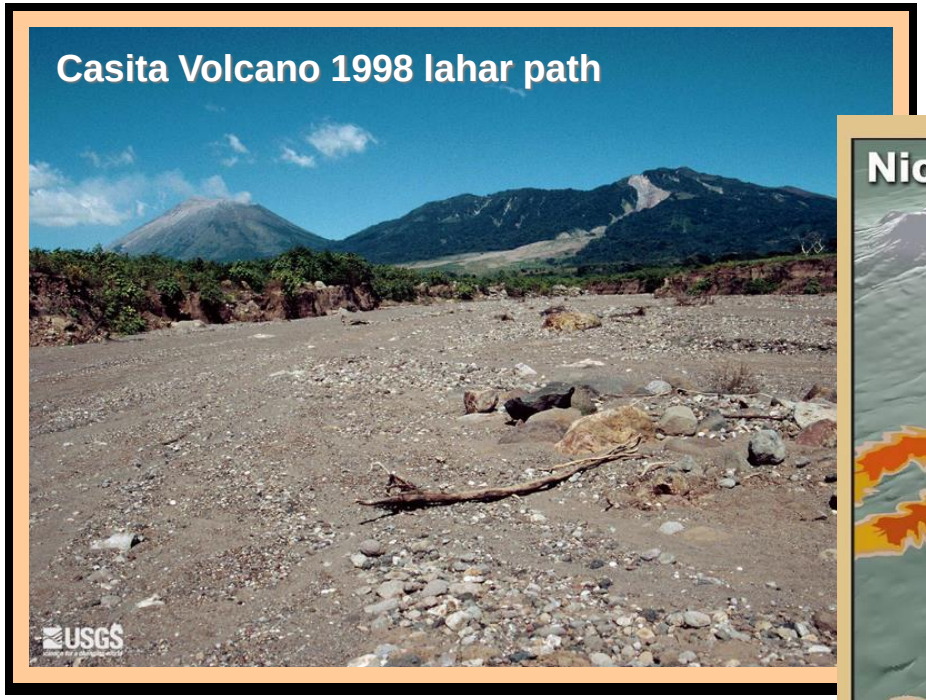
Statistical models
(for example, LAHARZ for lahars, energy cone for PDCs)

- Use statistical/empirical models like LAHARZ (for lahars) or energy cones (for PDCs) for rapid assessment and portrayal of hazard zones
- Especially good for broad regions (such as an entire volcano) where considerable uncertainty exists about the locations and sizes of future flows.
- Can run statistical models for the best-fit regression and the upper and lower confidence bounds to show the range of uncertainty (this is now a built-in option in LAHARZ) to reflect the effects of natural variability

Physical models
(for example, DCLAW for lahars, TITAN2D for PDCs)

- Use physically-based models (like DCLAW) to perform more detailed hazard assessments where prior knowledge warrants, and/or the model has input parameters that are more easily estimated (TITAN2D).
- Such models have great flexibility and explanatory power, but they also have more components—and more sources of uncertainty—than a simple statistical model such as LAHARZ or energy cones.
- A physically based model like DCLAW such can have excellent explanatory power but still have limited predictive power. Other physical models with fewer input parameters (like TITAN2D for PDCs) are more easily used for forecasting.
- Thus, they must be used carefully if they're used in a hazard forecasting mode. One model run ≠ hazard map.

Forecasting lahar inundation limits using LAHARZ

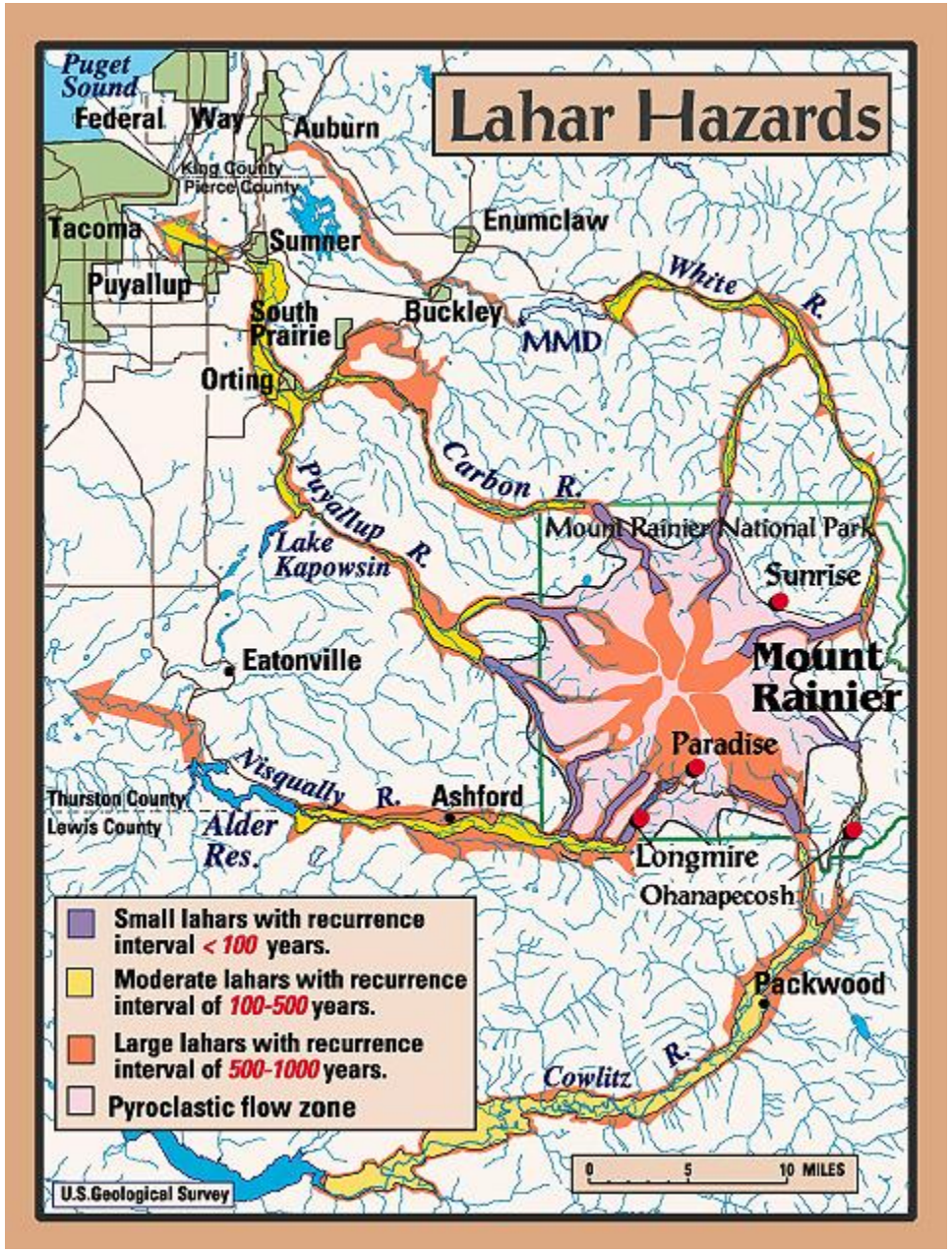




Acknowledgements

- Many slides presented here are courtesy of Steve Schilling and Julie Griswold
- Further reading:
  - Schilling. 1998. LAHARZ: GIS programs for automated mapping of lahar-inundation hazard zones. USGS Open-File Report 98-638
  - Iverson et al. 1998. Objective delineation of lahar-inundation hazard zones. Geological Society of America Bulletin, 110, 972-984
  - Griswold & Iverson 2007. Mobility Statistics and Automated Hazard Mapping for Debris Flows and Rock Avalanches. USGS Scientific Investigations Report 2007–5276 Version 1.1
  - Schilling. 2014. Laharz_py: GIS Tools for Automated Mapping of Lahar Inundation Hazard Zones. USGS Open-File Report 2014–1073

- Objective, reproducible?
- Map details send the wrong message?
- What are we communicating?
- Automated methods?
- Access to remote areas?
- Geologic history/maps completed?
- Time during crisis for field work?

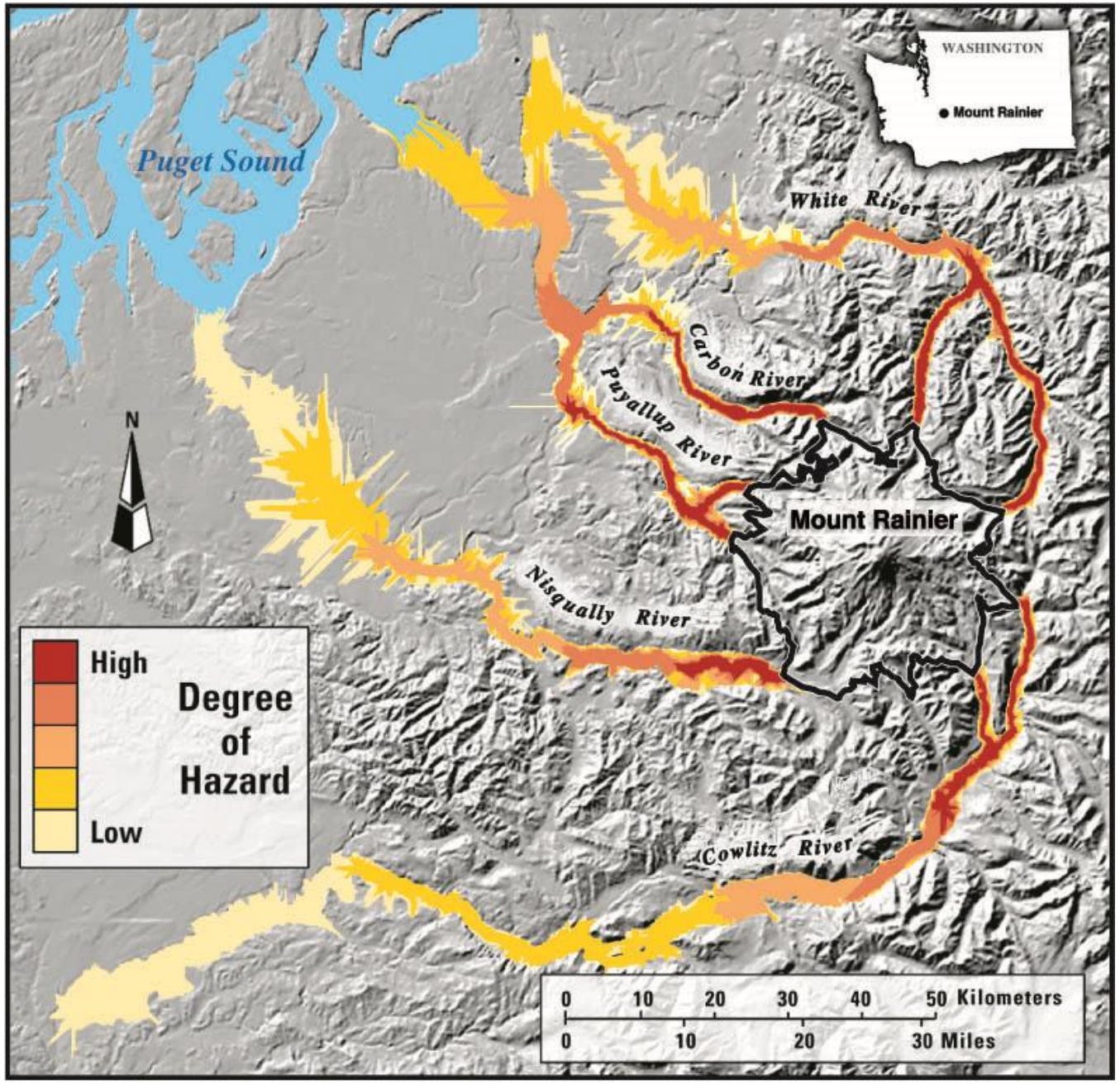


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LAHARZ

Goal: a rapid, objective, reproducible method for forecasting lateral and distal inundation limits that define hazard zones

Guiding Philosophy: Global data on areas inundated by past lahars provide a basis for forecasting future inundation limits at particular locations



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LAHARZ principles

1) inundation by past lahars provides a basis for predicting inundation by future lahars

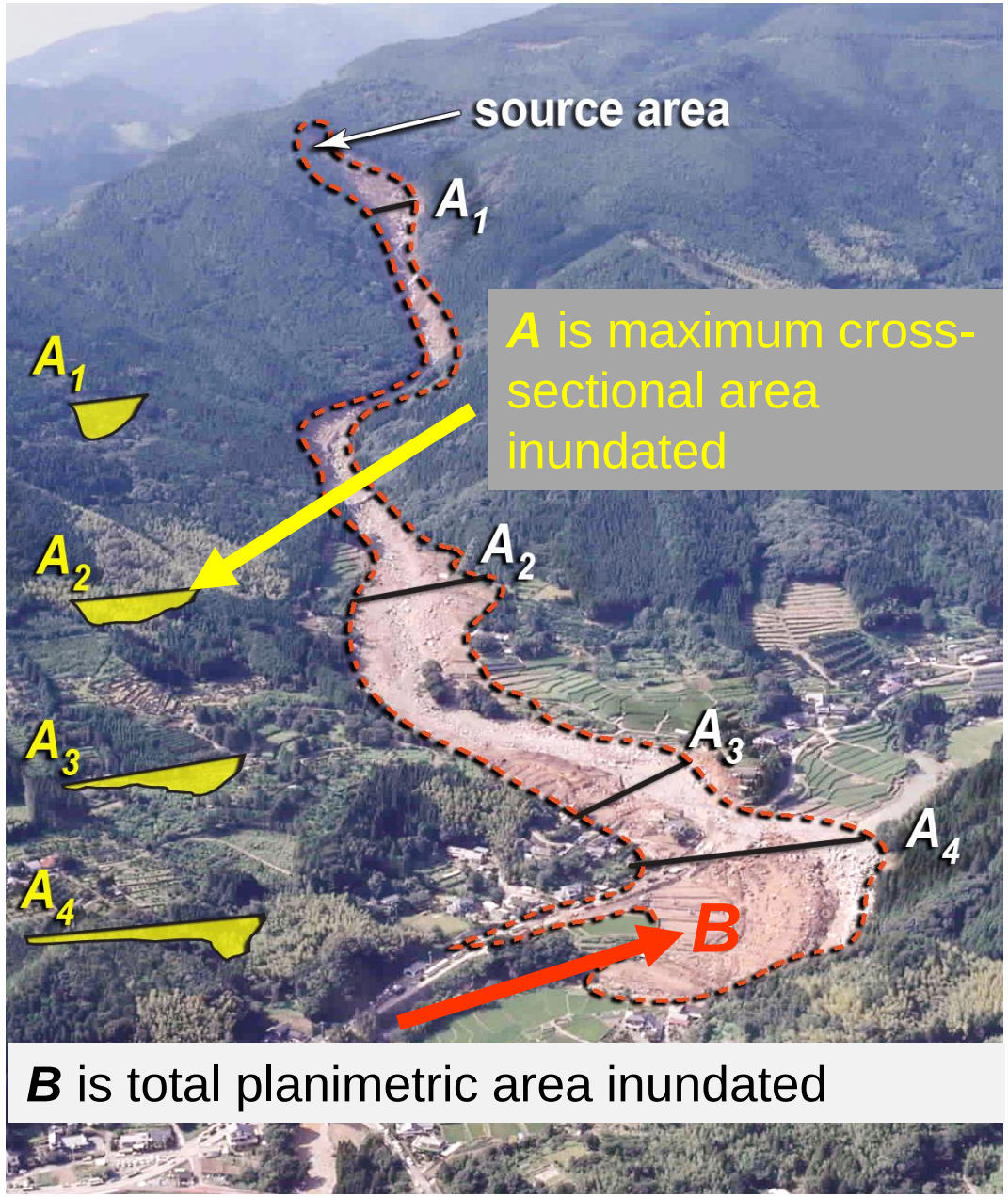
2) distal lahar hazards are confined to valleys that start on volcano flanks

3) lahar volume largely controls the extent of inundation downstream

4) voluminous lahars occur less often than small lahars

5) no one can foretell the exact size of the next lahar to descend a given drainage

(Iverson et al., 1998).

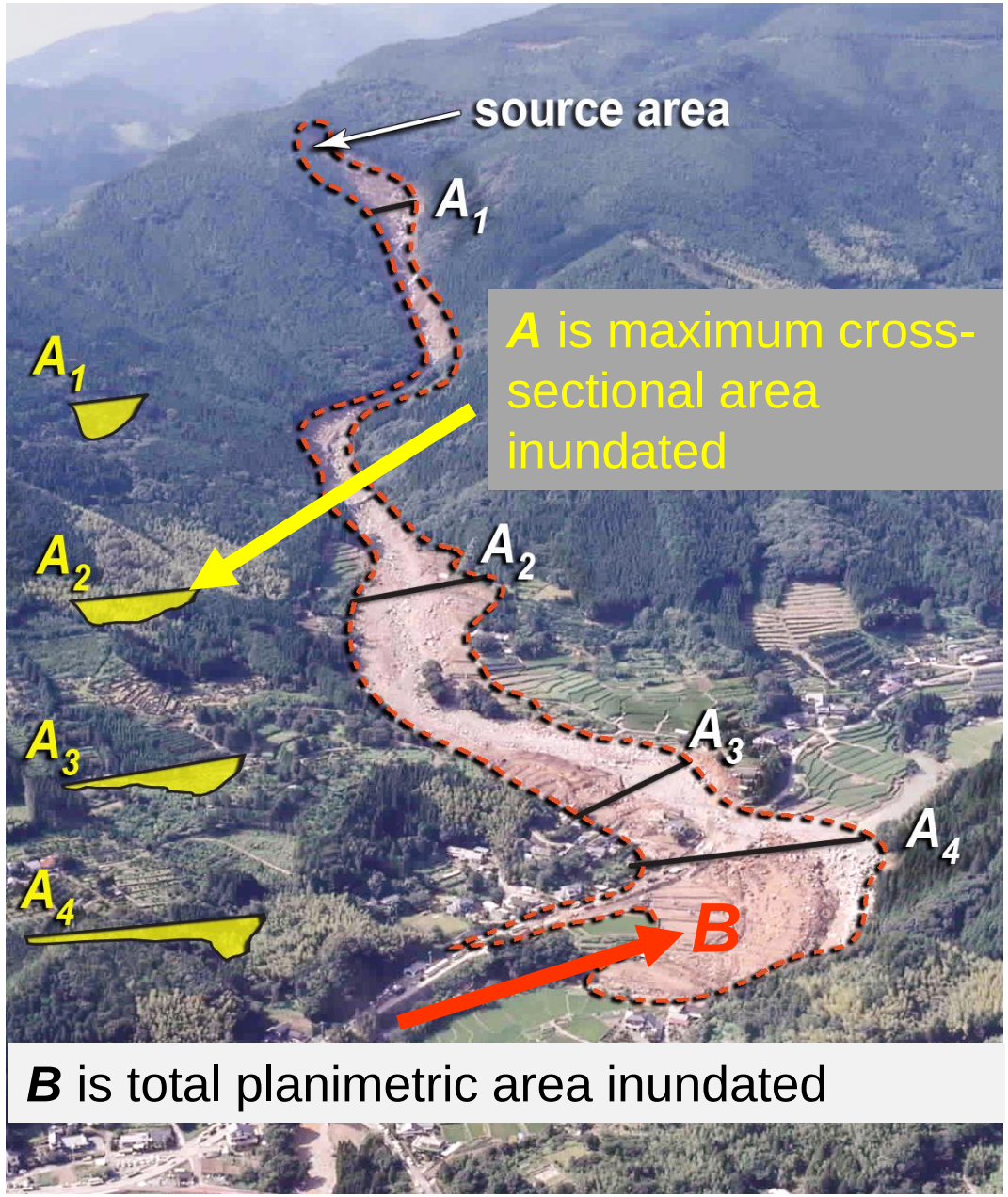


LAHARZ principles

Any flow advancing down-valley produces lateral and distal limits in other words:

cross-sectional (A) and planimetric (B) inundation areas that are influenced by topography and flow volume, *V.*

LAHARZ considers these two influences (topography and volume). The method uses three variables only: Volume (V), cross-sectional area (A), and planimetric area (B).



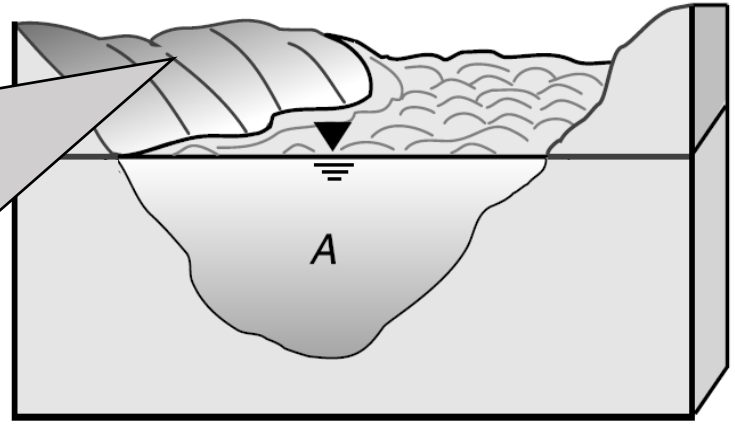
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LAHARZ equations: Use past events to forecast future events

Cross-sectional and planimetric area of inundation are proportional to flow volume$^{2/3}$

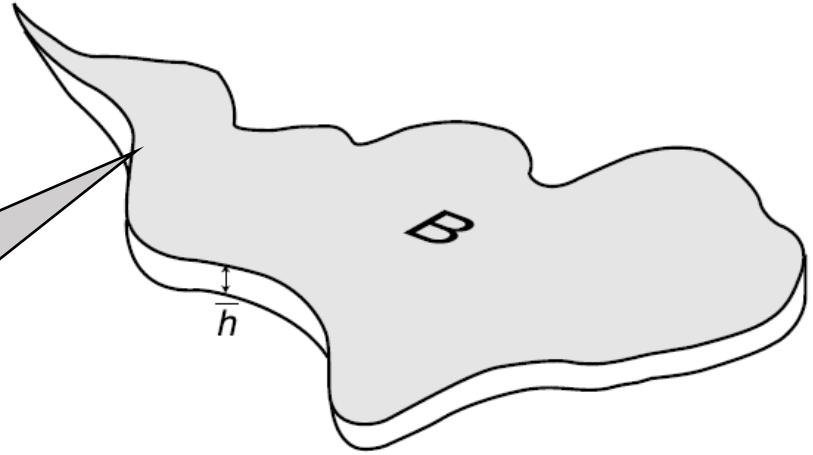
The coefficients c_1 and c_2 are determined from measuring the maximum cross-sectional area, planimetric area, and volume of many flows and performing regression analysis

high-flow marks indicated by strandlines, levees, embedded gravels, or the height of stripped bark on the trunks of adjacent trees, and the height of log jams.



Max cross-sectional area (A) = $c_1 V^{2/3}$

lateral limits of any evidence of high-flow lines.



Total planimetric area = $c_2 V^{2/3}$

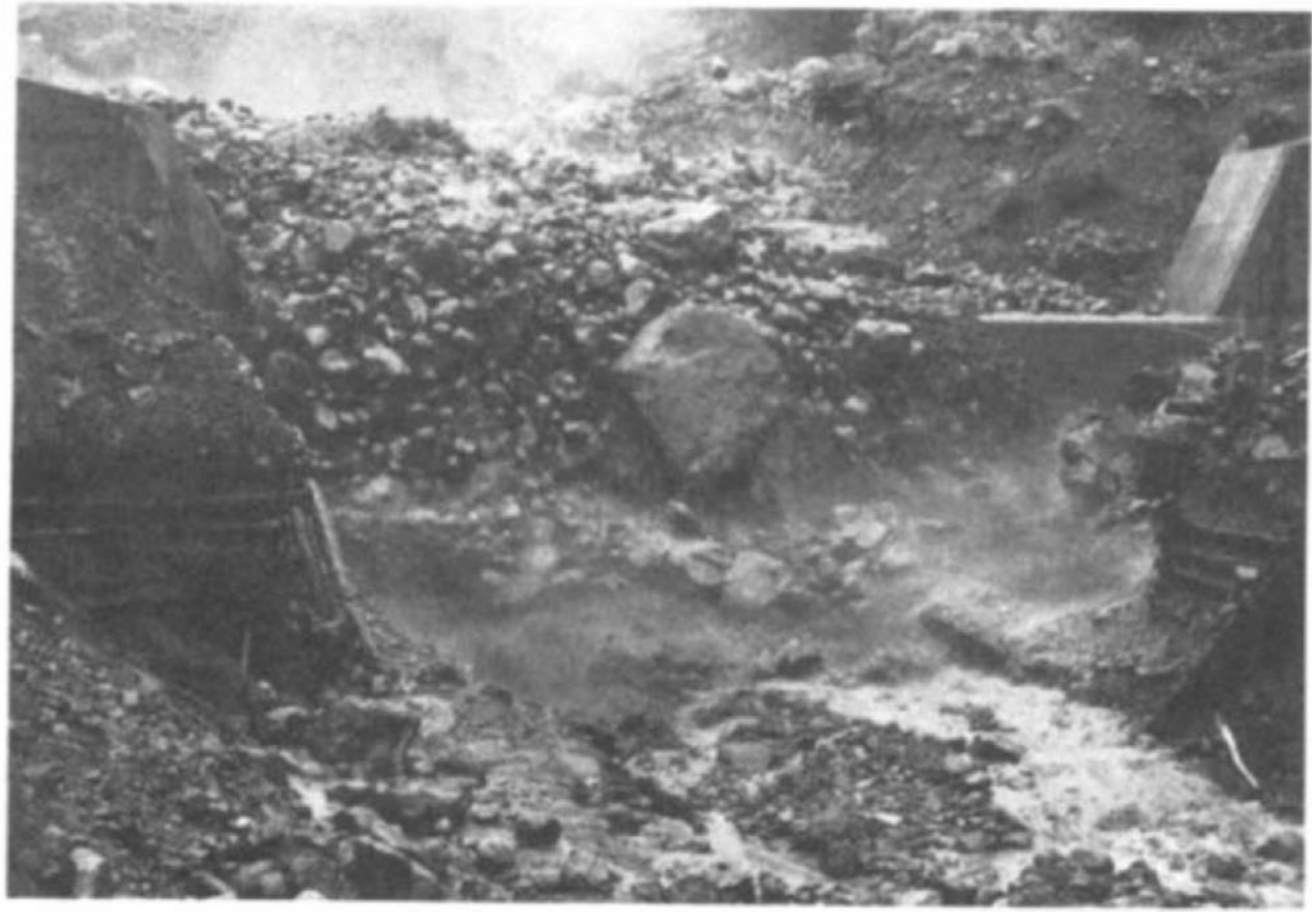
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Takahashi, 1981



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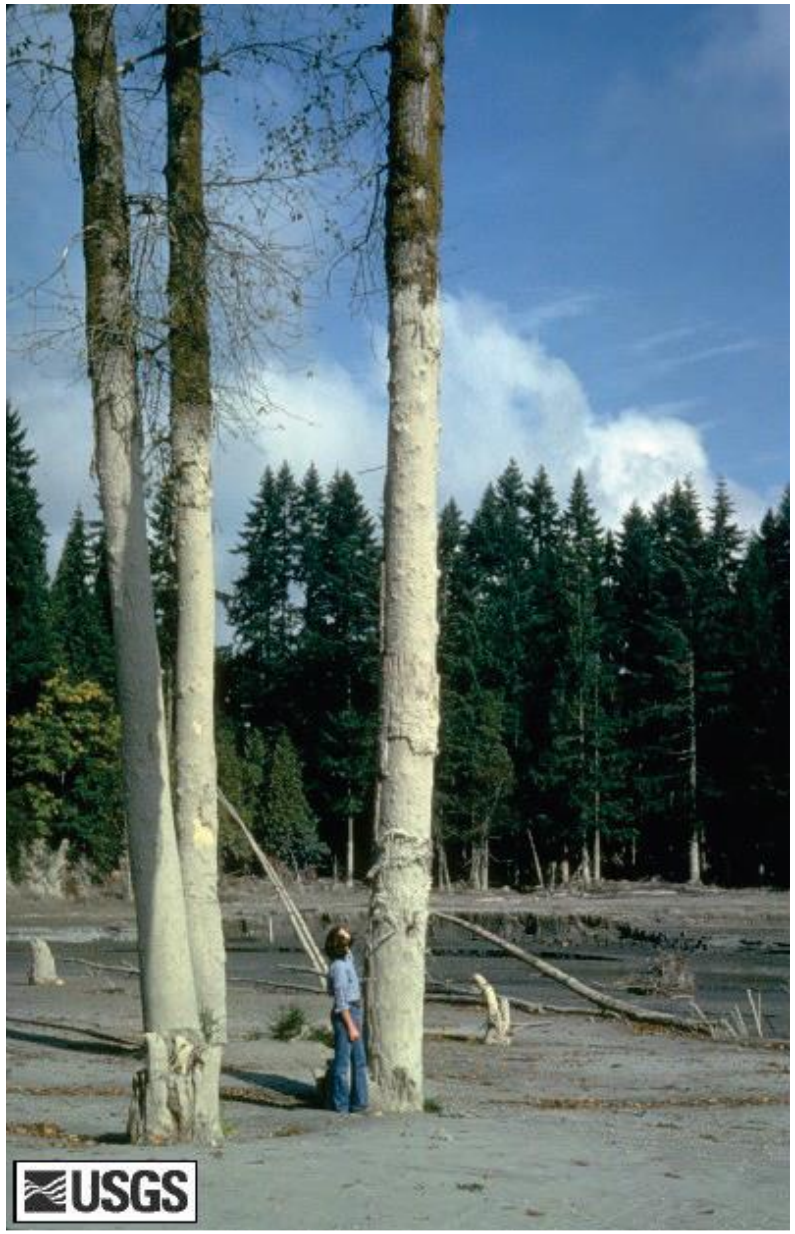


Lahar Deposit- Mount St. Helens

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May 18, 1980 eruption of Mount St. Helens

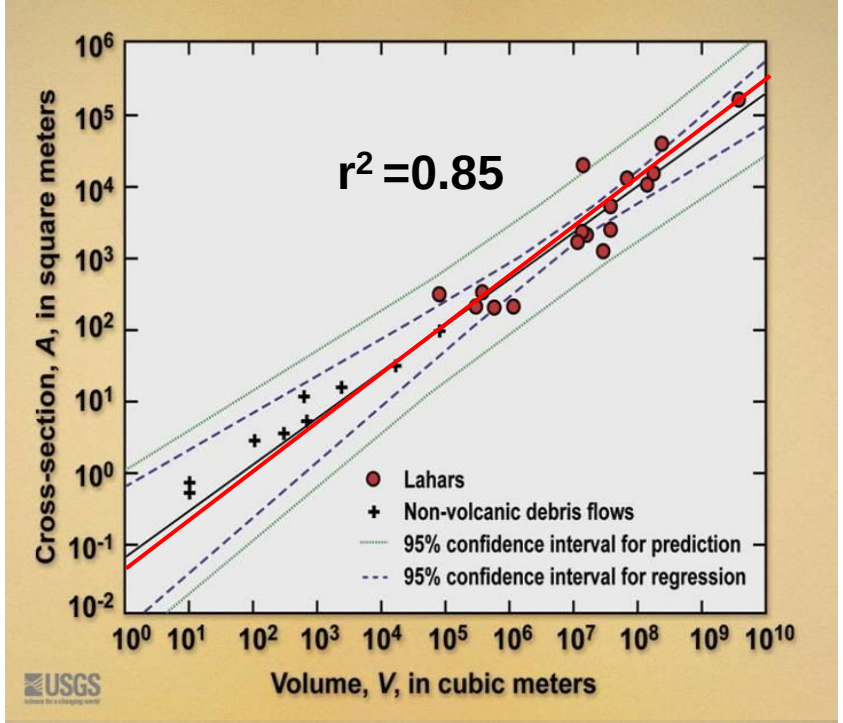
Photograph taken by Lyn Topinka, U.S. Geological Survey, 1980

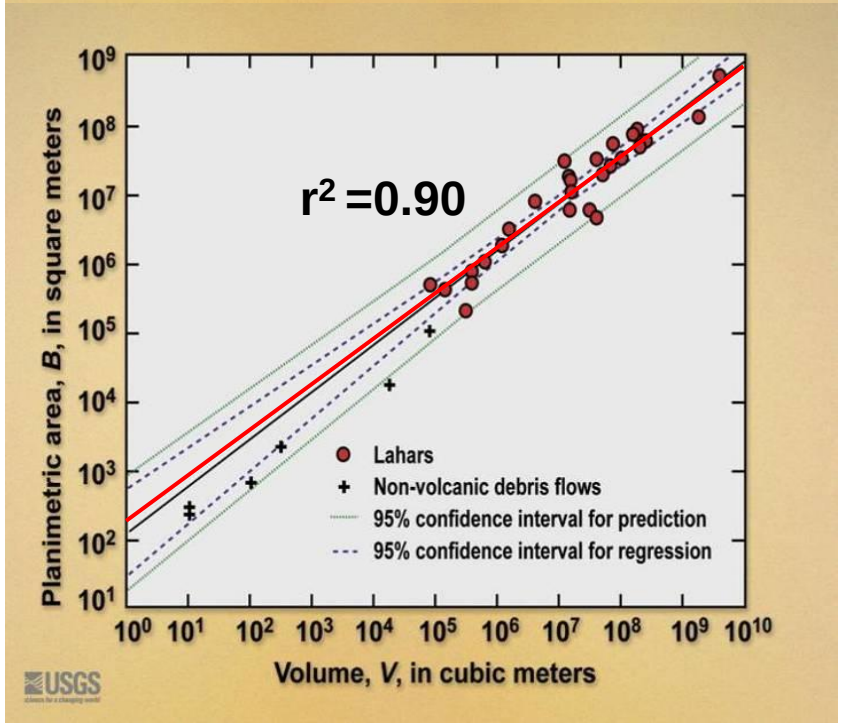
LAHARZ equations: Use past events to forecast future events

Cross-sectional and planimetric area of inundation are proportional to flow volume$^{2/3}$

The coefficients c_1 and c_2 are determined from measuring the maximum cross-sectional area, planimetric area, and volume of many flows and performing regression analysis

Different flow types (lahars, rockfalls, debris flows) have different coefficients





Cross-sectional area = $c_1 V^{2/3}$

Planimetric area = $c_2 V^{2/3}$

Flow Type	C_1	C_2
Lahars	0.05	200
Debris flows	0.1	20
Rockfalls	0.2	20

Some key points about the LAHARZ equations:

- ***Cross-sectional area* = 0.05 *V*** $^{2/3}$ and ***planimetric area* = 200 *V*** $^{2/3}$
- The calibrated values 0.05 and 200 reflect a specific data set
  - in this case a group of 27 large lahars worldwide.

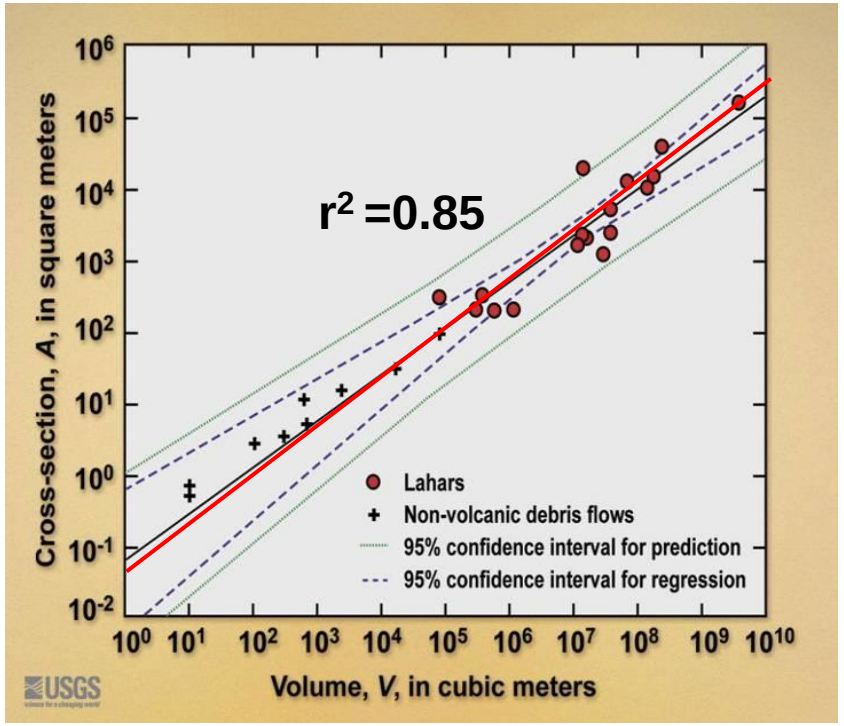
- NOT deposit volume and area, rather maximum flow volume, inundation area, and maximum cross-sectional inundated area as determined from high-flow marks, hydrographs, flow observations, etc.

- Values of the coefficients are specified to only one significant digit.

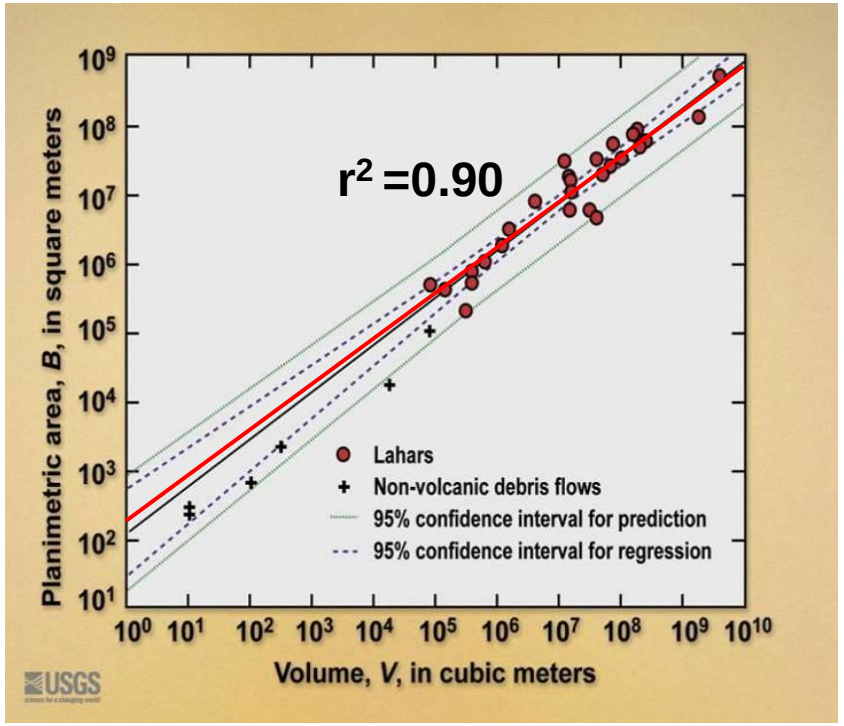
- Predictions based on the equations have statistical uncertainty portrayed by confidence intervals

- Error magnitudes are assumed to scale with *V* magnitudes --an assumption implicit in use of logarithmically transformed data and regression equations.

How does LAHARZ work?



***Cross-sectional area* = 0.05 $V^{2/3}$**

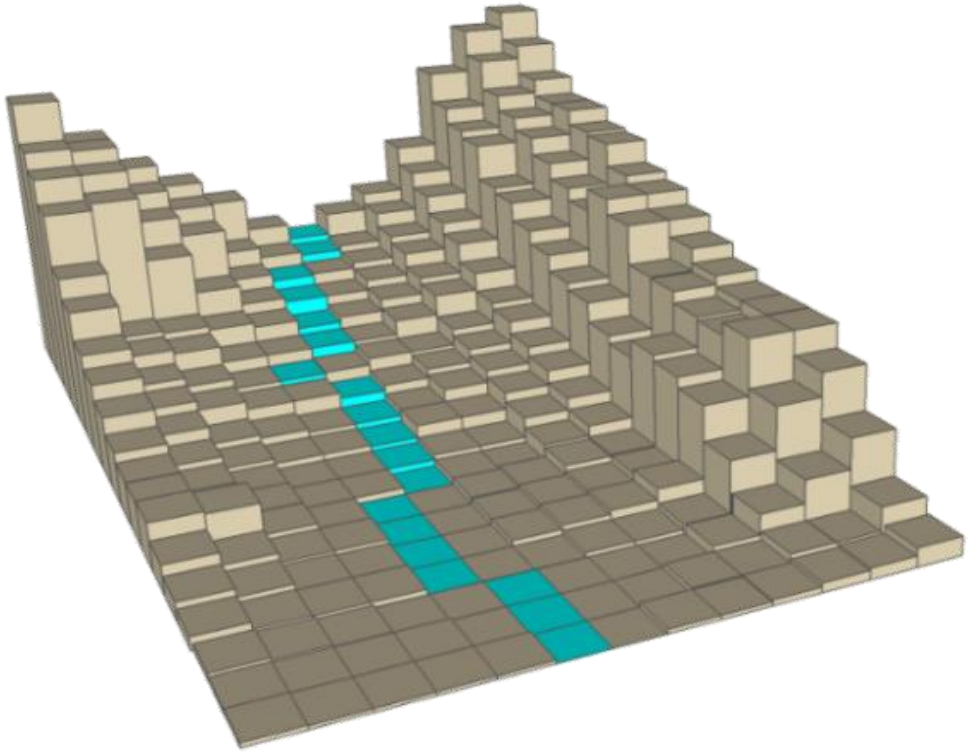


***Planimetric area* = 200 $V^{2/3}$**

From collected deposit data, determine the coefficients $\mathbf{c_1}$ and $\mathbf{c_2}$ for
Cross-sectional area = $\mathbf{c_1}V^{2/3}$
Planimetric Area = $\mathbf{c_2}V^{2/3}$

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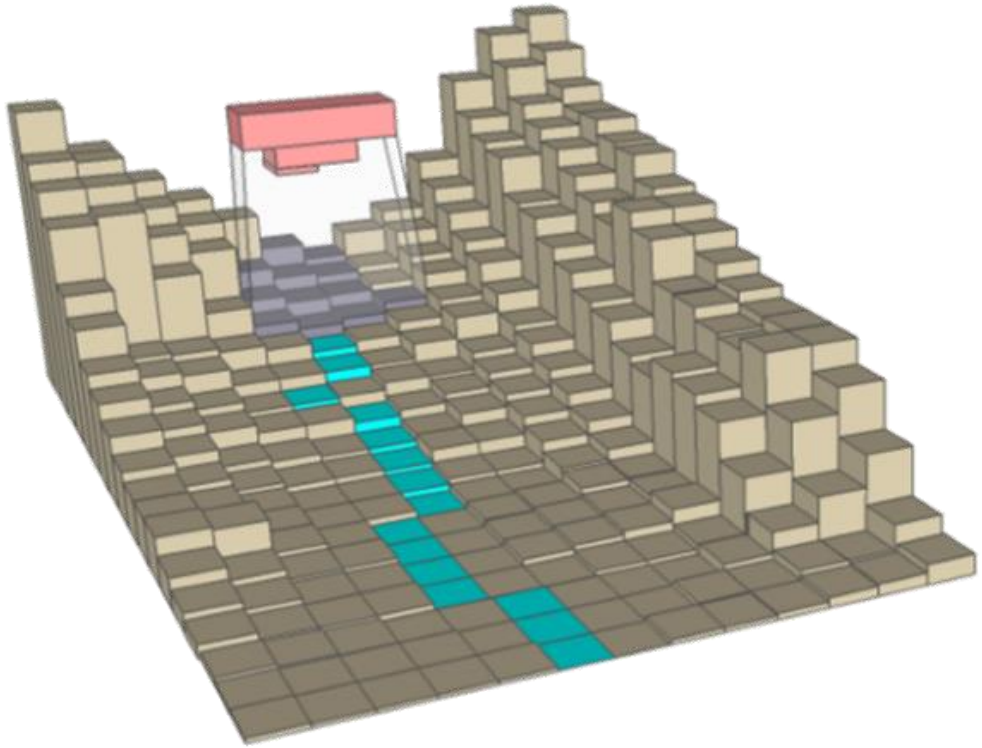
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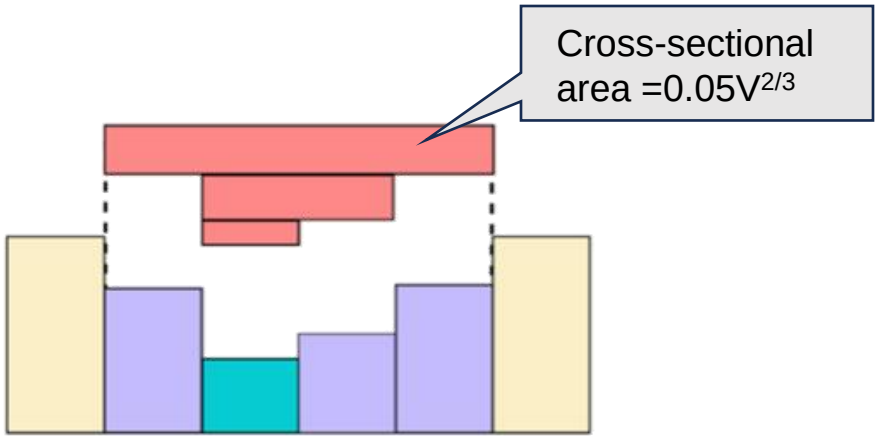


Use LAHARZ to fill sinks in topography, create a hillshade, and produce a stream network of valley thalweg cells.

Specify the volume, V, used for the simulation.

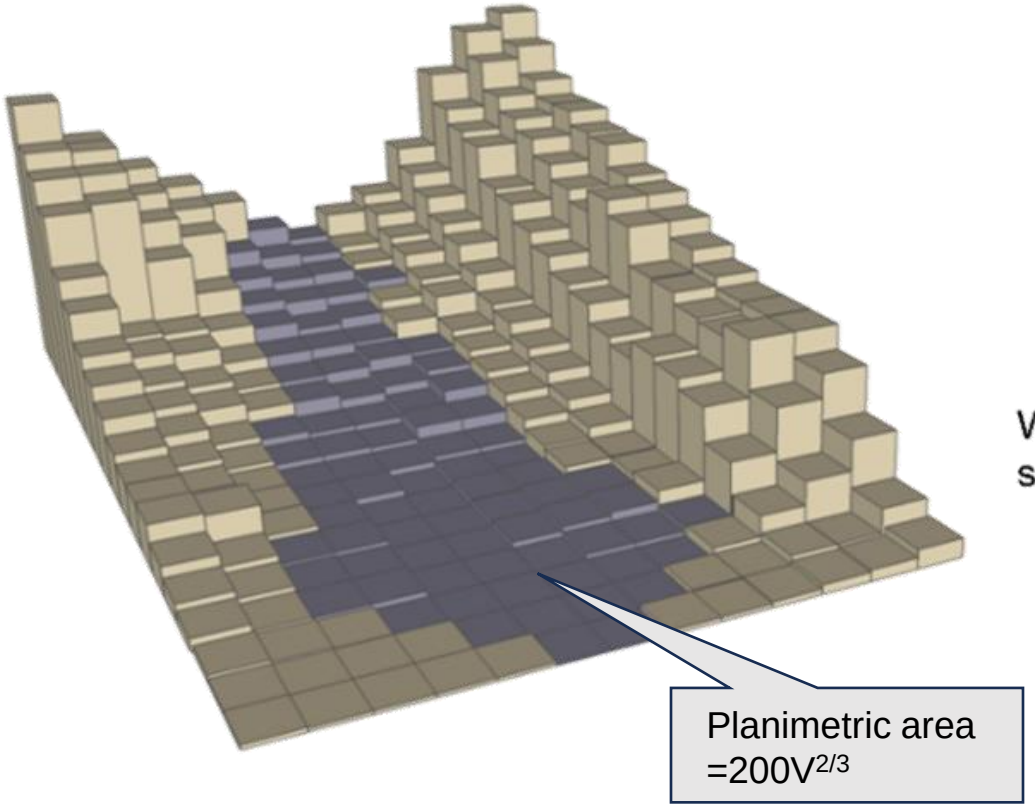
How does LAHARZ work?





Starting with the first specified valley thalweg cell, fill topography perpendicular to the thalweg until cross-sectional area is equal to $C_1V^{2/3}$. Add inundated cells to running planimetric area total, and move to next valley thalweg cell.

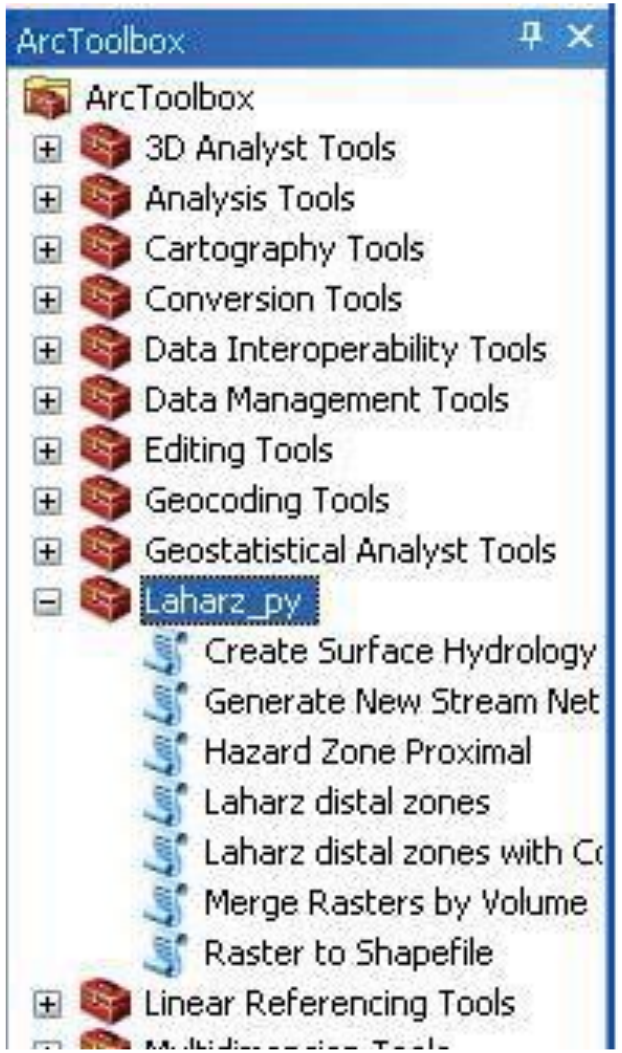
How does LAHARZ work?



When the running planimetric area total is equal to $C_2V^{2/3}$ stop the simulation.

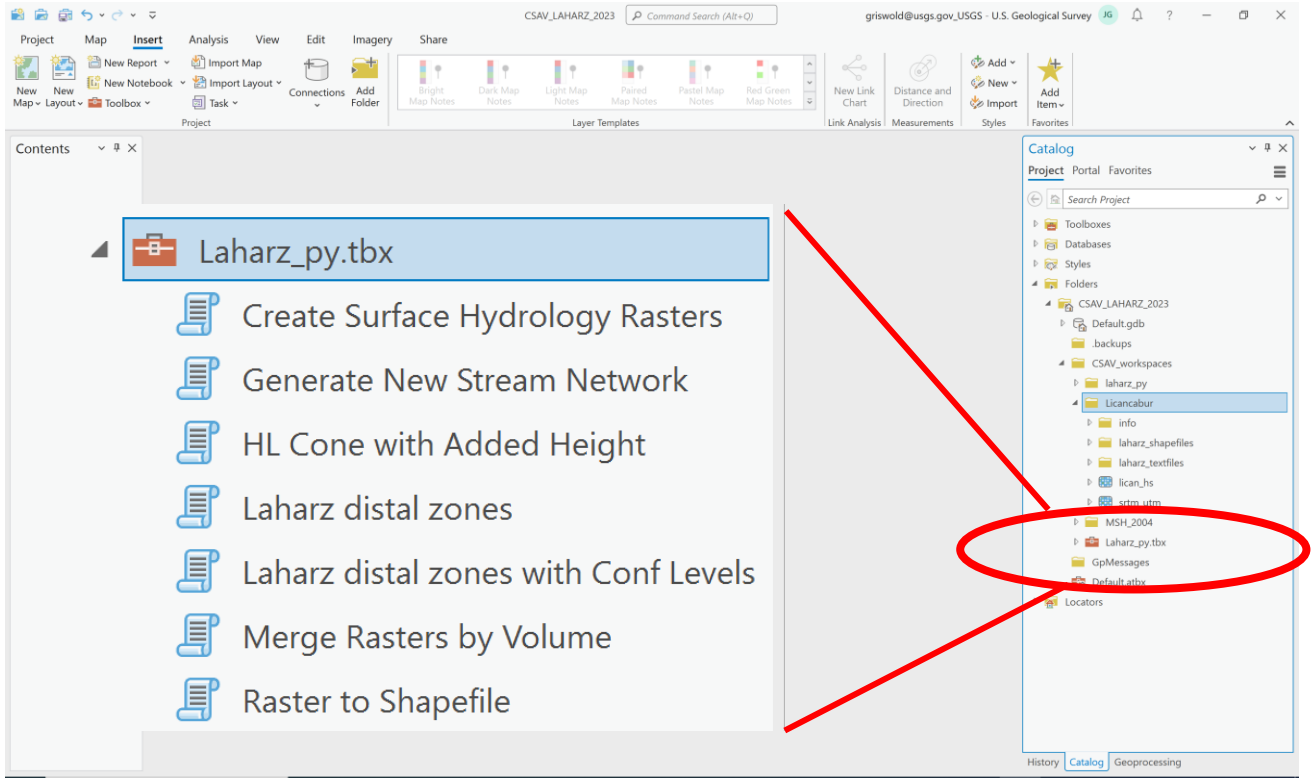
What do you need to run LAHARZ?

- ArcGIS ArcMap 10.x
- Laharz_py ArcGIS toolbox
- Schilling et al. 2014
- https://pubs.usgs.gov/of/2014/1073/
- CSAV file folders



What do you need to run LAHARZ?

- ArcGIS Pro
- Julie Griswold 2023 rewrite of Schilling et al. 2014 code for ArcGIS Pro
- Email me!
- CSAV file folders

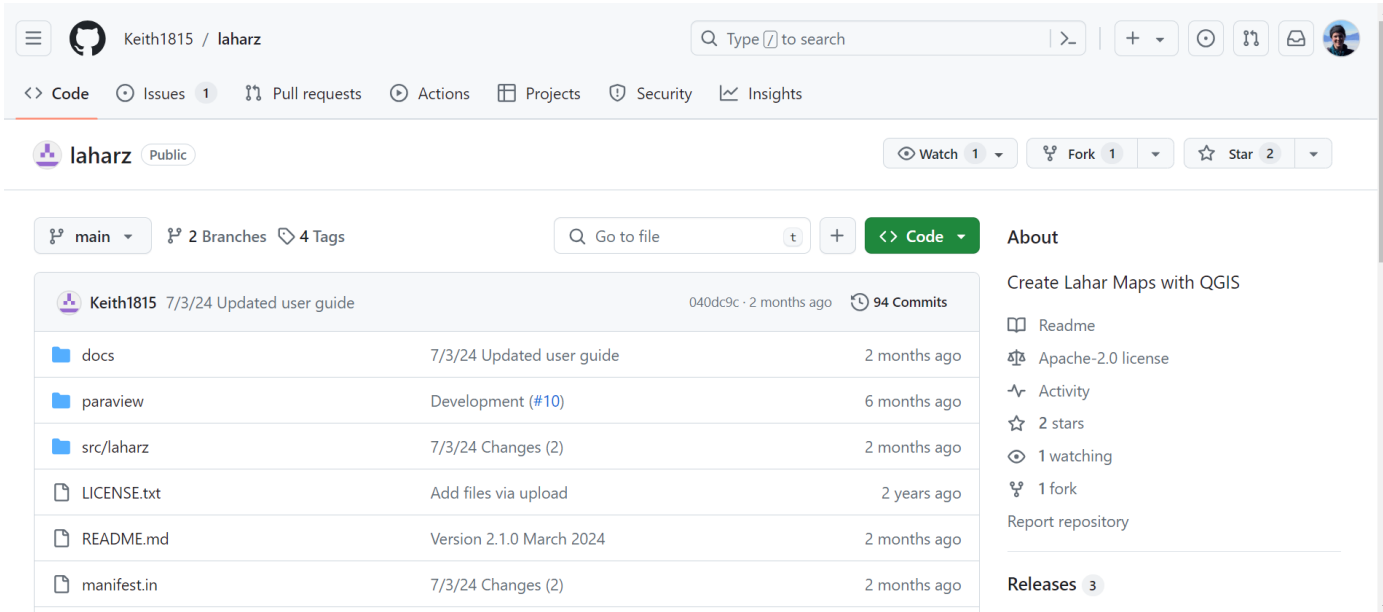


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What do you need to run LAHARZ?

- Keith Blair stand-alone python version
- https://github.com/Keith1815/laharz





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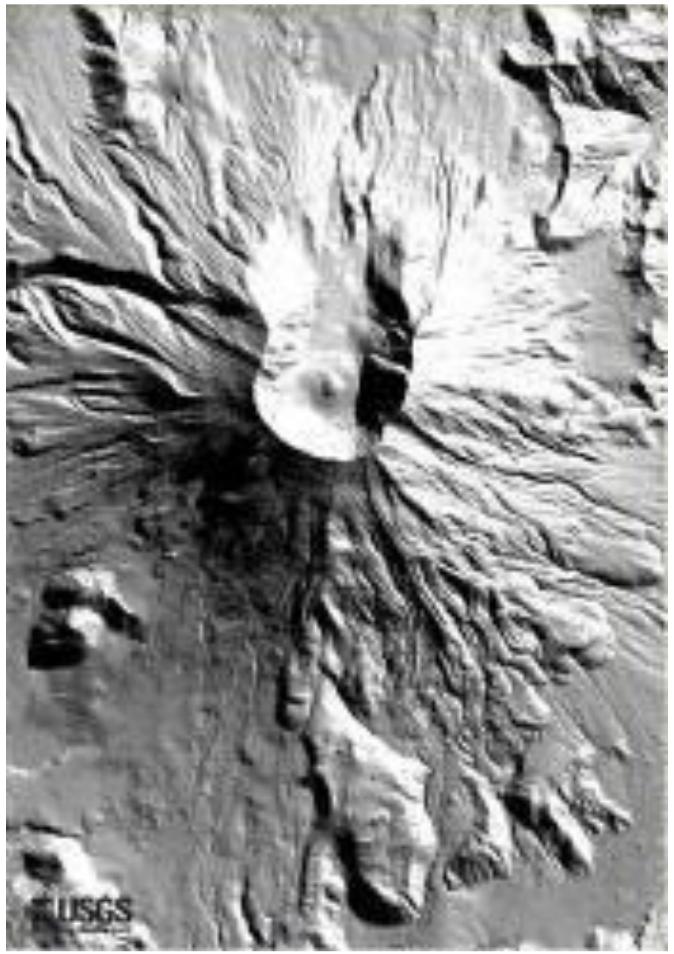
What do you need to run LAHARZ?

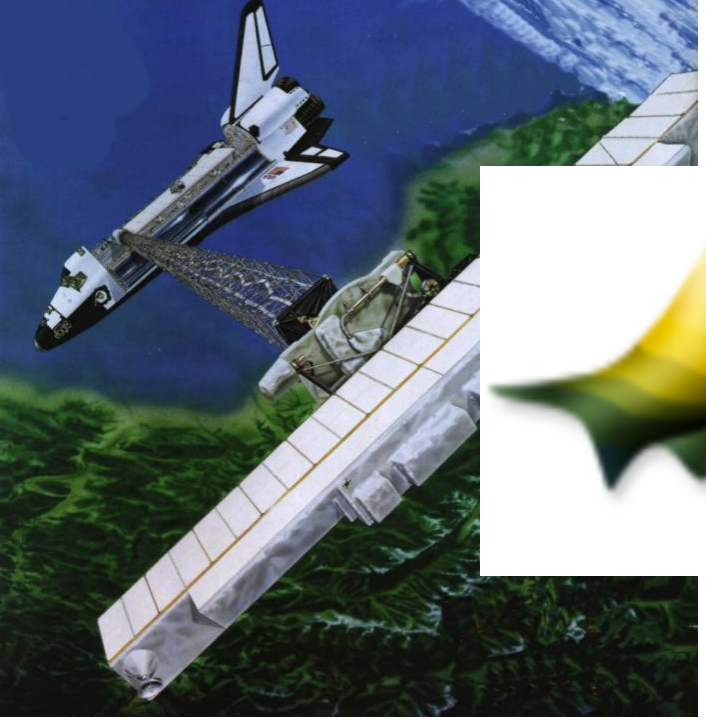
- VICTOR platform (using QGIS and python-version of LAHARZ by Keith Blair)
- https://victor.2i2c.cloud/
- Nice because you don't have to worry about installing python dependencies, so we'll use it for class



What do you need to run LAHARZ?

- A Digital Elevation Model (DEM) that has an adequate extent (includes the volcano and drainages) and that accurately represents the hydrology







OpenTopography

High-Resolution Topography Data and Tools



SRTM

Shuttle Radar Topography Mission

Survey for 11 days in February 2000

Radar “sees” through clouds

Data gaps: coastlines, mountains

60° N to 56° S

90-meter global dataset

Vertical error several meters

ASTER

Advanced Spaceborne Thermal Emission and Reflection Radiometer (imagery instrument)

1.3 million images 1999-2009

Data gaps: persistent clouds

83° N to 83° S

30-meter global dataset

Vertical error (20m) & greater due to limited ground control, especially mountains

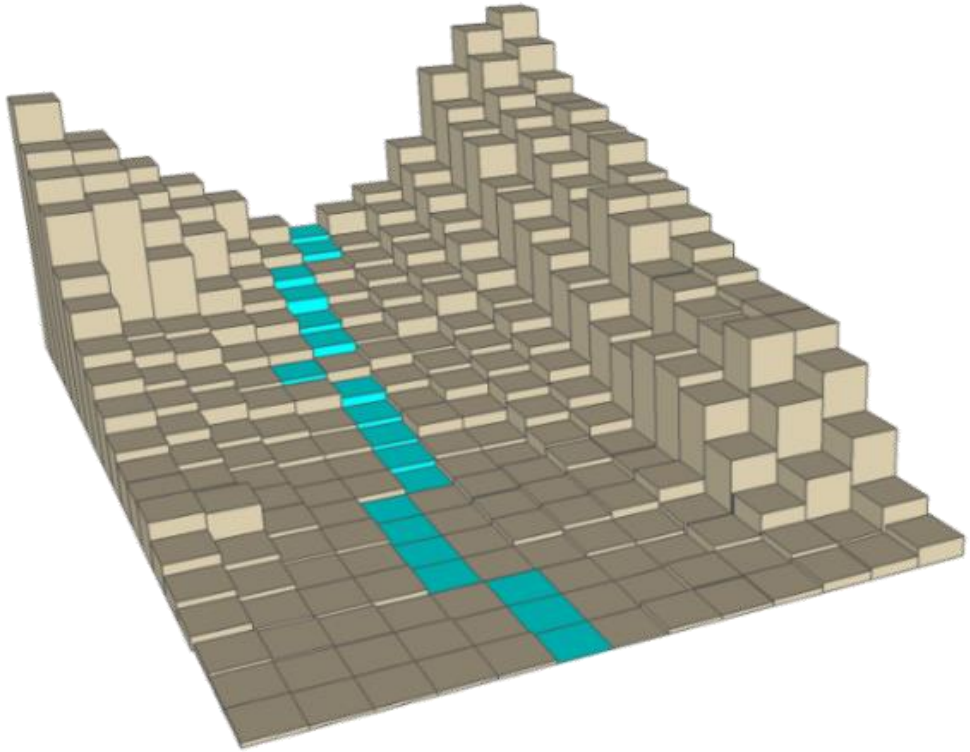
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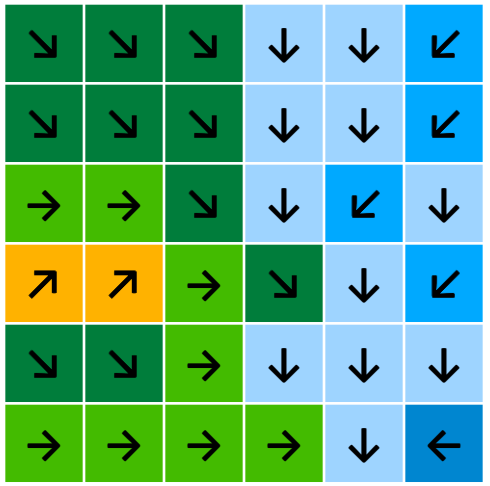
What do you need to run LAHARZ?

- A stream network
- “Create surface hydrology rasters” and “Generate Stream Network” tools in ArcGIS LAHARZ, the “r.stream.extract” in QGIS, or the newest version of the standalone Python LAHARZ
- Generates stream network using the DEM, and a user defined stream threshold
- This is the number of upstream cells that have to flow into a cell to consider it a stream cell. A smaller number here will make a stream network with more branches and longer streams.
- The network should generally match published maps, but should represent the *current* stream network



78	72	69	71	58	49
74	67	56	49	46	50
69	53	44	37	38	48
64	58	55	22	31	24
68	61	47	21	16	19
74	53	34	12	11	12

Elevation (DEM)



Flow direction

0	0	0	0	0	0
0	1	1	2	2	0
0	3	7	5	4	0
0	0	0	20	0	1
0	0	0	1	24	0
0	2	4	7	35	1

Flow accumulation

0	0	0	0	0	0
0	1	1	2	2	0
0	3	7	5	4	0
0	0	0	20	0	1
0	0	0	1	24	0
0	2	4	7	35	1

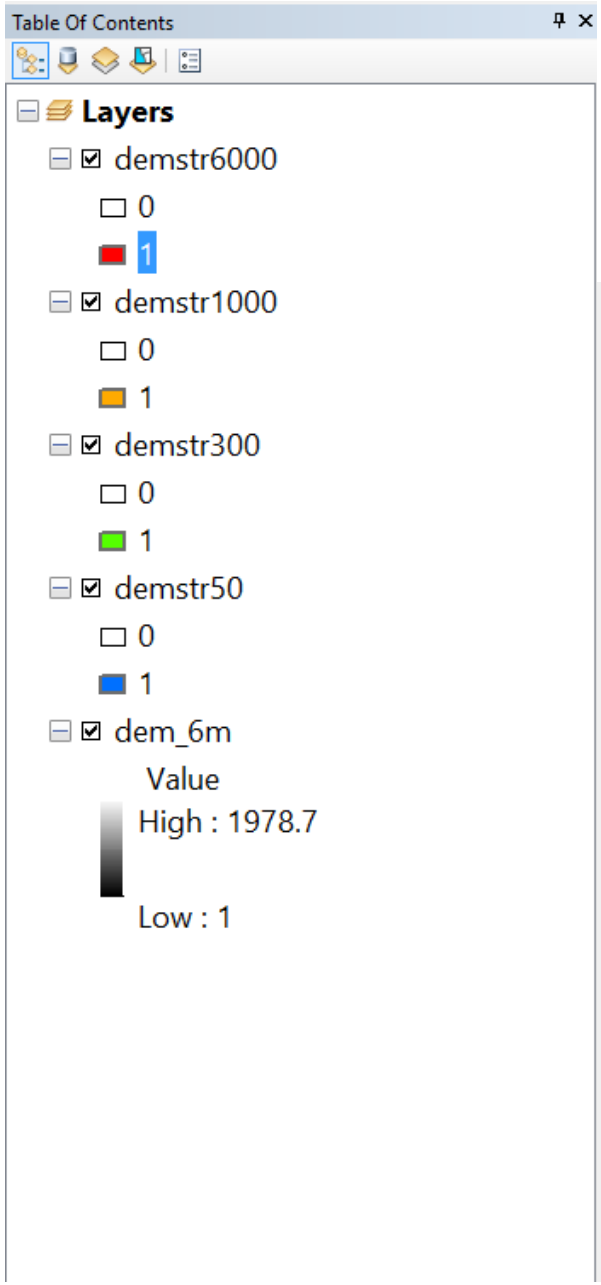
Stream threshold = 2

0	0	0	0	0	0
0	1	1	2	2	0
0	3	7	5	4	0
0	0	0	20	0	1
0	0	0	1	24	0
0	2	4	7	35	1

Stream threshold = 6

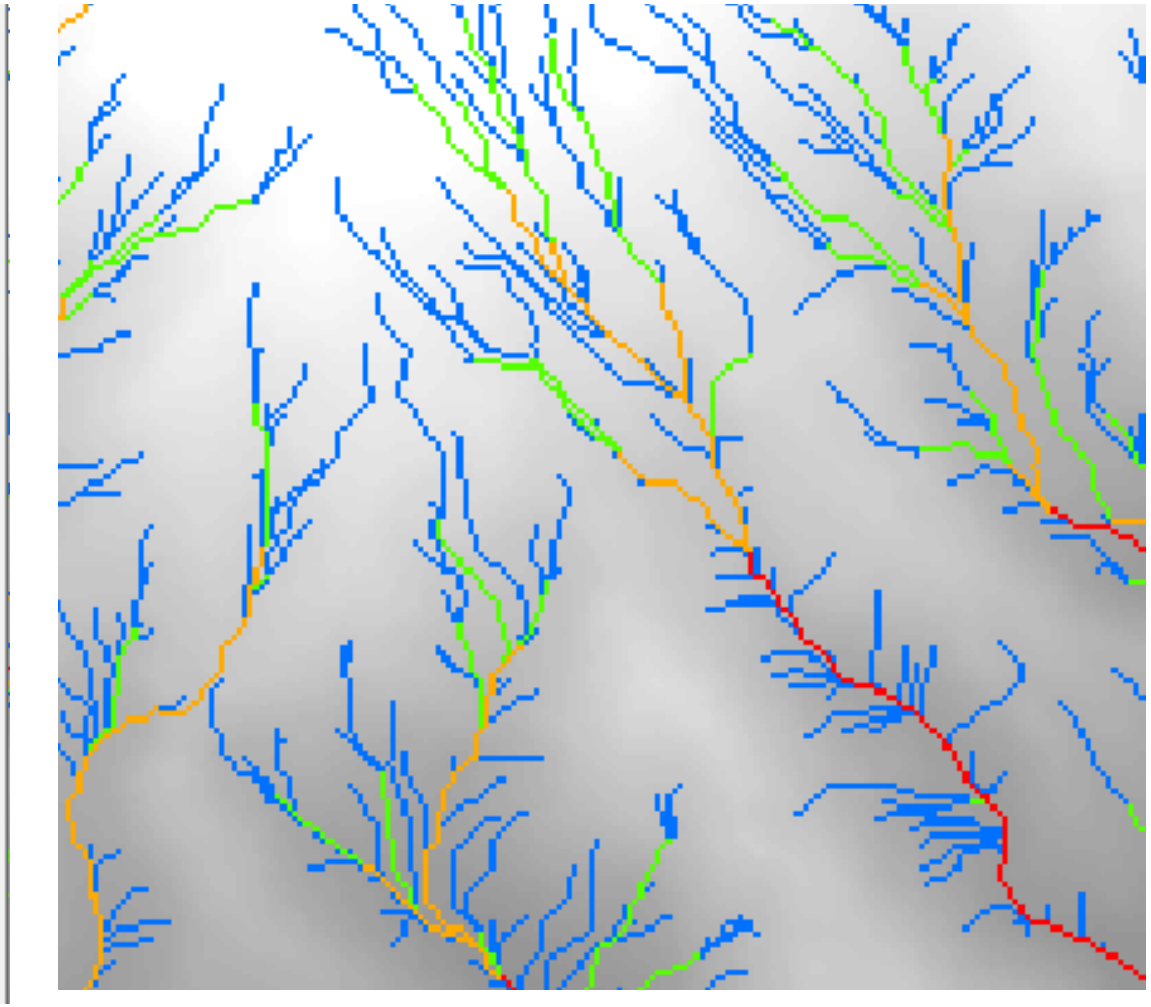
Lower stream threshold: longer stream network with more branches

Higher stream threshold: shorter stream network with fewer branches



High stream threshold: shorter stream network with fewer branches

Low stream threshold: longer stream network with more branches



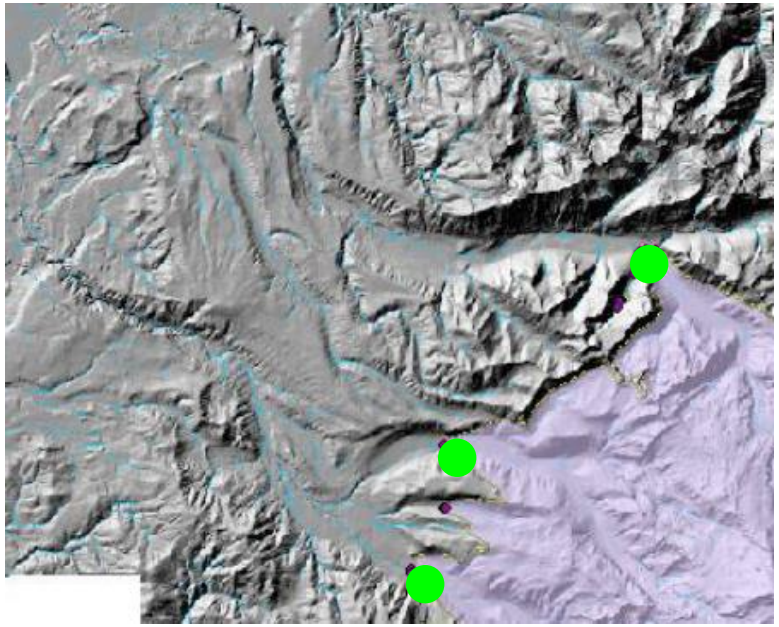
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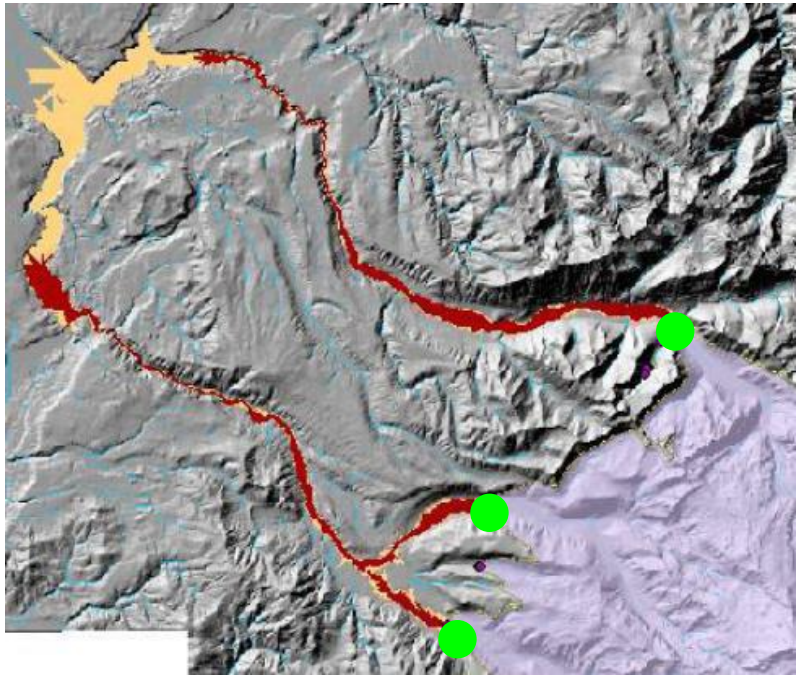
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What do you need to run LAHARZ?

- A starting location or locations:
  - Use the proximal hazard zone tool within LAHARZ to produce an H/L cone that defines the starting locations (can edit these to have fewer points)
  - Manually select starting locations

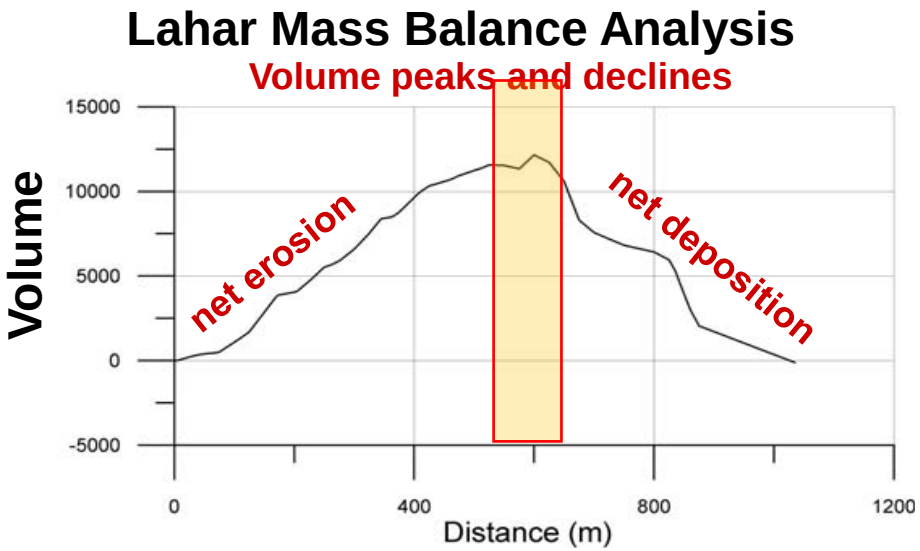




Proximal Hazard Zone for LAHARZ

Why use H/L Cone for lahars?

- Lahars may initiate anywhere on the volcano edifice. We can't know an exact start location in advance.
- Loose pyroclastic flow deposits and ashfall are often the source material for lahars and are most abundant within the H/L cone.
- Lahars gain volume (entrain material) on steep slopes.



- So, we start modeling lahars by estimating where on the flanks the flow volume is greatest.
- We do this by defining an H/L cone that marks the transition between erosion and deposition. That is, inside the H/L cone, the volume is still increasing. Outside the H/L cone, the volume is maintained or decreasing (depositing).
- How do we identify an H/L cone that represents this transition?
    - from net erosion to net deposition
    - from steep slopes to moderate slopes (20°-30°), that is, find the break in slope
    - from pyroclastic debris fans to more channelized flows

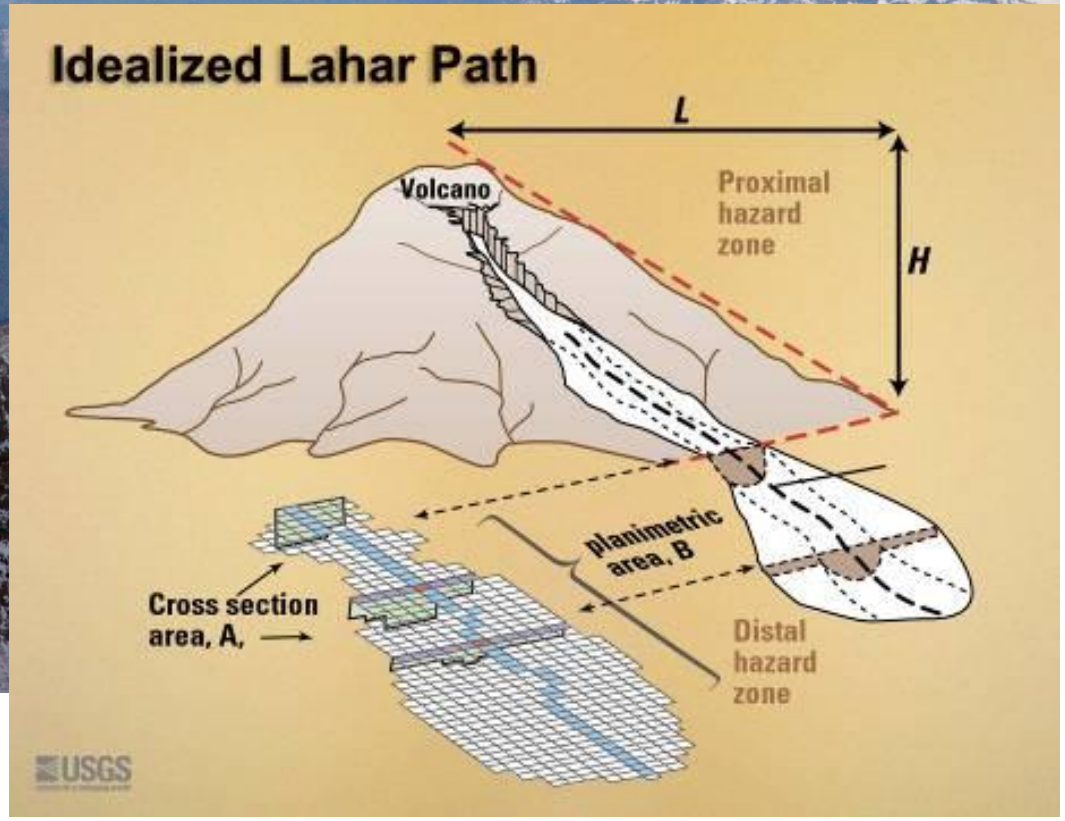
Mt. Rainier, west side

L

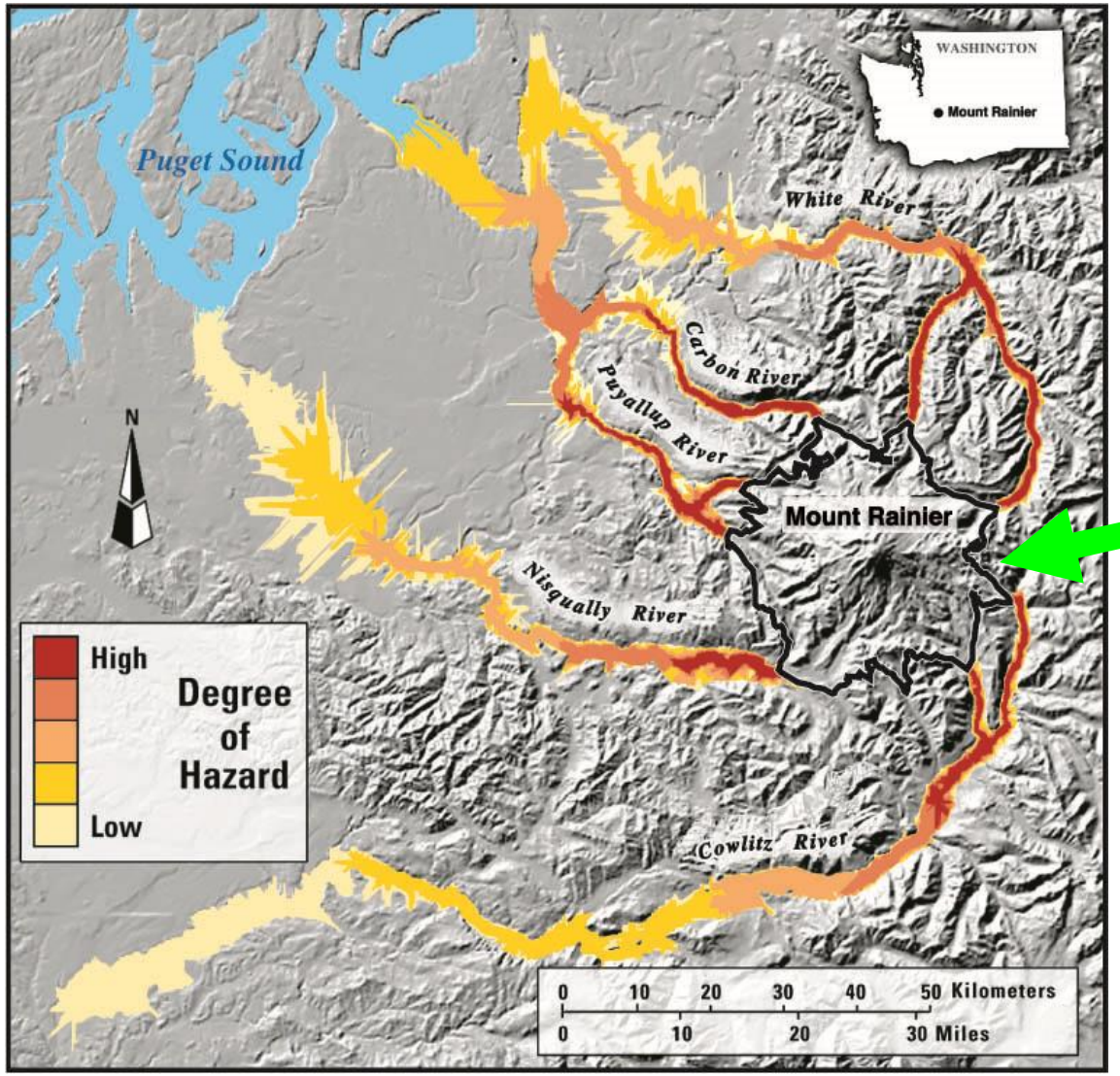
V

H

Begin flow hazard-zone computation where deposition first exceeds erosion



Lahar hazard zones computed for Mt. Rainier with proximal hazard zone defined by *H/L* = 0.23



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Special Cases: not a nice pointy volcano?



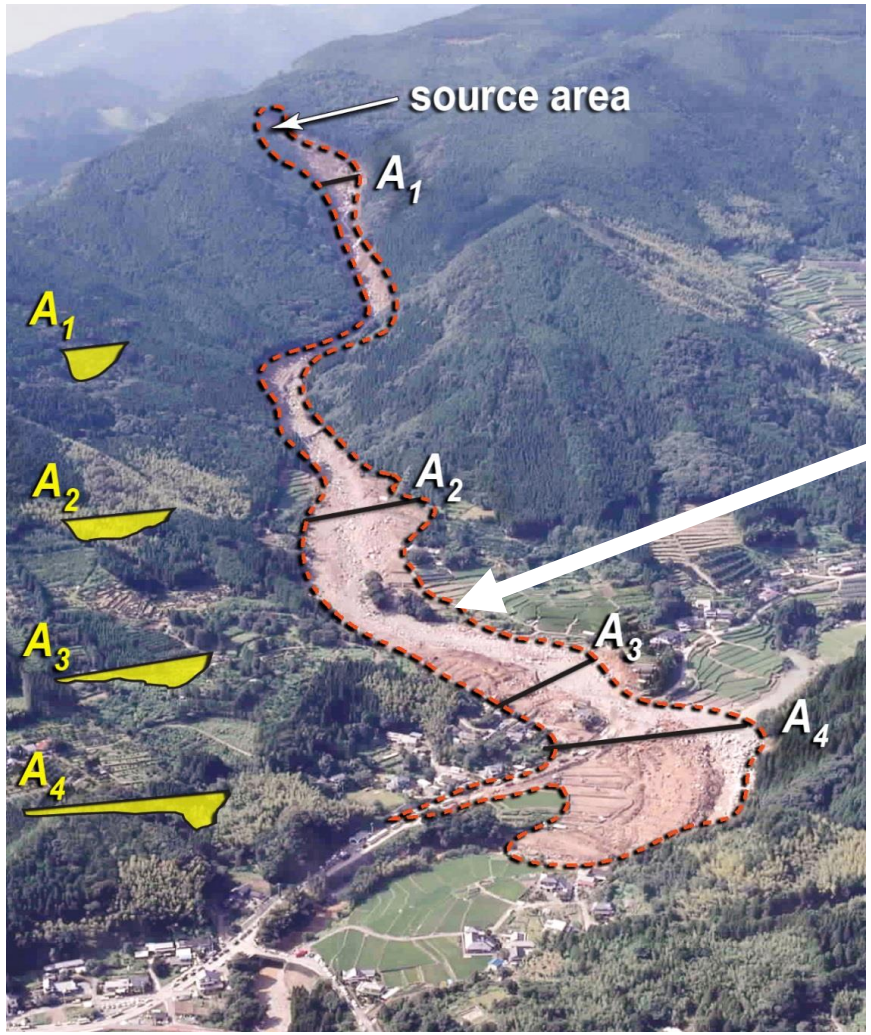
Special Cases: not a nice pointy volcano?
Add elevation in LAHARZ



What do you need to run LAHARZ?

- A reasonable range of lahar volumes, considering volcano size, type, history, geology and the goal of the forecast (worst-case scenario vs. most-probable scenario)
- Because of uncertainty and naturally occurring error in the regression equations, you should choose volumes that increase by at least 2x (1, 2, 4, 8, 16 million m^3, for example)
- Even better is to use values that increase by $10^{0.5}$ (1, 3, 10, 30, 100 million m^3, for example)

The flow volume *V* of interest in LAHARZ is the <u>maximum</u> volume attained along the flow path... that is, the volume where the flow transitions from net erosion to deposition.



Vmax is defined where net erosion changes to net deposition

In real events *V* can change and evolve along the flow path, but this evolution is not considered in LAHARZ. In LAHARZ the volume of interest is the **maximum volume** that occurs during the course of flow.

Summary of key points

1. Documented inundation limits and flow volumes of past lahars provide the data for statistical forecasting of lateral and distal inundation limits for future lahars.

2. Lahar flow volume is the independent variable that must be specified by the user. Generally it's best to posit a range of potential volumes and examine a range of possible outcomes.

3. A rationale must be adopted for establishing starting points for computing inundation limits. For volcanoes we typically use the intersection of *H/L* cones with valley thalwegs to establish starting points.

Hands-on session to try it out!